

Dawid Ciupka

DEVOPS / PLATFORM ENGINEER

I build systems, automate everything, and make infrastructure actually work.

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ABOUT

DevOps / Platform Engineer with a background in Industrial Automation. I build and operate real infrastructure – self-hosted Kubernetes clusters, CI/CD pipelines, IoT integrations, and embedded systems. I work at the intersection of systems engineering and software, bridging the gap between hardware and cloud-native tooling.

Previously a PLC programmer (Allen Bradley, Siemens) and mechanical designer (Autodesk Inventor), now focused on Kubernetes, Docker, Python, MQTT and making infrastructure actually work.

Automate first

Build real things

Hands-on engineering

No bullshit, only working systems

PROFESSIONAL PROJECTS

Smart Building Platform

- **Context:** Client from the building management sector.
- **Scope:** Deployment and configuration on a self-hosted Kubernetes cluster managed via Rancher. Full platform architecture design, TLS certificate management, and enterprise authentication integration.
- **Stack:** Kubernetes · Rancher · Docker · SAML
- **Result:** Production-ready smart building platform with secure SSO authentication and containerized deployment.

Environment Monitoring System

- **Context:** Client from the pharmaceutical sector (confidential).
- **Scope:** Architecture design and deployment of an environment monitoring platform on Kubernetes. Handled full PKI and certificate lifecycle, implemented OAuth 2.0 and SAML-based SSO for enterprise identity provider integration.
- **Stack:** Kubernetes · Rancher · OAuth 2.0 ·
- **Result:** Secure, compliant monitoring platform integrated with client's enterprise identity infrastructure.

PERSONAL PROJECTS

Kubernetes Cluster (Rancher & HA)

- **Problem:** Need for a production-grade, highly available local environment to orchestrate microservices and IoT workloads without recurring cloud costs.
- **Solution:** Designed and deployed a multi-node K8s cluster using Rancher for management. Implemented MetallB for Load Balancing, Longhorn for distributed block storage, and Traefik as Ingress Controller. Integrated Prometheus and Grafana for full-stack observability.
- **Stack:** Kubernetes · Rancher · Longhorn · MetallB · Prometheus · Grafana · Docker
- **Result:** A resilient, self-healing infrastructure hosting 20+ services. Successfully automated deployment workflows via CI/CD, achieving 99.9% uptime for local services.

Smart Shutters System

- **Problem:** Manual window shutters – no scheduling, no remote control, no smart home integration.
- **Solution:** Fullstack app with Django REST backend and JavaScript frontend. Physical shutters controlled via MQTT messages to embedded devices.
- **Stack:** Django · MQTT · JavaScript · Raspberry Pi · Docker
- **Result:** End-to-end home automation: web UI + physical device control + scheduling. Fully self-hosted.

AI Camera (Raspberry Pi)

- **Problem:** Need for real-time AI image processing at the edge, without relying on cloud APIs.
- **Solution:** USB camera emulator that intercepts the video stream, applies AI transformations on-device, and outputs a processed feed – transparent to applications.
- **Stack:** Python · Raspberry Pi · OpenCV · Linux
- **Result:** Edge AI processing with no cloud dependency. Low-latency real-time video transformation.

Hardware Random Generator (Teensy)

- **Problem:** Software PRNGs are not truly random – insufficient for cryptographic or security-sensitive use cases.
- **Solution:** Microcontroller-based entropy source using hardware noise. Data exposed via USB interface (migrating to Ethernet) and consumed by backend services.
- **Stack:** Teensy · C++ · Python · USB/Ethernet
- **Result:** True hardware entropy source integrated with software backend.

SKILLS

INFRASTRUCTURE

Kubernetes
Rancher
Docker
Terraform

CI/CD & AUTOMATION

GitHub Actions
Bash scripting
Linux

DEVELOPMENT

Python
Django
REST API
PostgreSQL

NETWORKING

DNS, TCP/IP
Reverse Proxy
MQTT

EMBEDDED / IOT

Raspberry Pi
Teensy
MQTT

INDUSTRIAL

PLC (Allen
Bradley)
Siemens TIA
Portal
Autodesk
Inventor

EXPERIENCE

DevOps / Platform Engineer

2022 – Present

Self-employed / Freelance

- Built and operated self-hosted Kubernetes cluster (Rancher)
- Developed IoT and home automation systems (Django, MQTT)
- Implemented CI/CD pipelines with GitHub Actions
- Integrated embedded hardware (Raspberry Pi, Teensy) with backend services

Industrial Automation Engineer

2018 – 2022

– (add company name) –

- Programmed PLCs: Allen Bradley (Studio 5000), Siemens (TIA Portal)
- Mechanical design and documentation in Autodesk Inventor
- Commissioning and troubleshooting of industrial automation systems
- HMI development and SCADA integration

LANGUAGES

Polish – Native

English – Professional